

MADHVIKA SEHGAL

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PROFESSIONAL SUMMARY

Business Intelligence Engineer with 4+ years of experience across Amazon, Barclays, and Accenture, building data pipelines, analytics frameworks, and decision-support solutions across financial, operational, and customer domains. Strong in SQL, Python, AWS, data modeling, BI, and experimentation, with a track record of taking ambiguous problems from definition through validated recommendations. Experienced in modernizing reporting systems, improving data quality at scale, automating manual workflows, and translating complex data into decisions.

EDUCATION

Master of Science, Management Information Systems — University of Arizona, Tucson, AZ | Aug 2019 – Dec 2020

Bachelor of Technology, Information Technology — Guru Gobind Singh Indraprastha University, New Delhi, India | Aug 2012 – May 2016

TECHNICAL SKILLS

Languages & Query: SQL, Python (Pandas, NumPy), Spark SQL, SAS, Advanced Excel, Jupyter

Data Warehousing & Modeling: Data Modeling, Dimensional Modeling, Data Marts, ETL/ELT, Data Reconciliation, Data Quality Testing

Cloud & Data Platforms: AWS (Redshift, Athena, S3, Glue, Lambda), Oracle

BI & Visualization: Tableau, Tableau Prep, Amazon QuickSight, Alteryx

Analytics: A/B Testing, Hypothesis Testing, Funnel Analysis, Customer Journey Analysis, KPI Development, Statistical Analysis

Data Engineering & Workflow: DAG-based Pipelines, Git, Amazon internal tooling analogous to dbt/Airflow

Generative AI: Amazon Bedrock, Claude, LLM-assisted Analytics & Workflow Automation

PROFESSIONAL EXPERIENCE

Business Intelligence Engineer | Amazon Health – Pharmacy Finance & Analytics

Seattle, WA | Aug 2025 – Present

- Redesigned cost allocation in Amazon Pharmacy's Contribution Profit platform, replacing broad proxies with granular operational drivers; corrected a distorted pharmacist-vs-fulfillment cost-to-fill split by switching from blended payroll estimates to time-clock actuals, preserved GL/P&L reconciliation, supported Small Format's growth from 4 to 38 sites, and carved kiosk into its own flagged P&L.
- Remediated data-quality risk during a primary-key and schema migration, updating 4–5 downstream SQL jobs and adding validation guardrails across 10+ dependent datasets; preserved reporting continuity with zero material metric regression.
- Automated monthly workforce-cost allocation in SQL, integrating ADP payroll, warehouse metadata, operational flags, and configurable business rules to replace manual Excel formulas and pivots; saved 7+ hours per month and improved month-end reporting reliability.
- Developed a governance and adoption dashboard from 1M+ monthly platform audit events, classifying usage across datasets, users, and tools; gave leaders their first prioritized view of underutilized assets, drove decommissioning of 14 stale jobs and optimization of the 4 highest-compute jobs (~1 hour/day saved), and became the team's standard governance review.
- Implemented a Claude-assisted SQL investigation workflow for month-end GL and cost-allocation anomalies, using AI to generate root-cause hypotheses and targeted validation queries; independently verified findings through source reconciliation and regression checks, cutting debugging by ~3 hours per issue.
- Built and continuously refined an AI-assisted on-call workflow, converting recurring incidents into reusable SOPs and an ownership/escalation map; reduced issue-routing and triage overhead by ~4 hours per week across the rotation.

Business Intelligence Analyst | Barclays – Credit Card, Contact Center Analytics

Whippany, NJ | Jan 2021 – Jul 2025

- Designed the analytics data layer for the Avaya/Oracle-to-AWS Connect migration, reconciling disparate call schemas across S3, Athena, Redshift, and Oracle and standardizing containment, self-service, and operational KPI logic for 30+ Tableau dashboards; restored call-data completeness from 2.5M to 4.5M calls/month, exposing staffing gaps that initiated the downstream Callback program.
- Led a 3-member analytics workstream for the Contact Center Callback rollout, mapping legacy and callback call flows, redefining SLA/KPI calculations, modifying upstream SQL logic, and validating 45+ downstream dashboards and SAS reports; contributed to 12% higher customer retention and 8% fewer transfers.
- Engineered a serverless GenAI reporting workflow using Python, AWS Lambda, S3, Boto3, Amazon Bedrock, and Claude to transform structured report metadata into standardized lineage, KPI definitions, validation rules, and audit documentation, reducing a days-long documentation process to minutes.
- Created the first analytics framework for ~200K monthly IVR Open Question calls, defining standardized metrics for agent-to-agent transfers, sentiment, complaints, and Simple-vs-Complex queue performance; surfaced recurring complaint and sentiment patterns and became the standard monthly view Product and Operations used to prioritize routing improvements.
- Designed and ran the Simple-vs-Complex routing experiment as a live 2-month production A/B test (5K control, 5K treatment) evaluated with a two-proportion hypothesis test; validated a transfer-rate reduction from 40% to 36% (10% relative) and rolled it to production, cutting ~8K agent transfers per month.
- Performed customer-journey and funnel analysis across authentication, self-service, routing, and servicing events, reconstructing customer paths using SQL event sequencing and session-level logic to identify friction points and inform demand forecasting, workforce planning, and targeted IVR flow redesigns.
- Developed an agent-performance scoring framework after finding CSAT/NPS covered only ~10% of agents; added a full-coverage sentiment signal and combined sentiment, CSAT, and NPS into a weighted, Z-score-normalized index, extending comparable scoring to the full agent base and becoming QA's standard for flagging outliers and prioritizing coaching.

Business Consultant | Accenture – SAP MM Consultant

Delhi, India | Jan 2017 – May 2019

- Led full-cycle SAP MM programs for global clients, translating ambiguous business needs into technical specifications and facilitating stakeholder workshops, UAT, issue resolution, and post-go-live support, achieving 95% user adoption.
- Developed BRDs and functional specifications; coordinated business and ABAP teams to deliver and validate scalable reports and enhancements aligned with business KPIs and change-management needs.